

Brett Tregoning

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I deliver measurable business value through **data science** and **machine learning engineering**. My skills go beyond the data science tech stack. I work with stakeholders from other domains to understand the problem from every angle and build a scalable solution. I understand and apply the cutting-edge data science and AI techniques so that I can address needs in a variety of situations.

Skills

Scientific: Statistical Methods, Machine Learning (ML), Bayesian Optimization, Adaptive Experimental Design (AE), Deep Learning, Neural Networks (NN), Transformers, Natural Language Processing (NLP), Large Language Models (LLM), Computer Vision (CV), Reinforcement Learning (RL), Decision Trees, Predictive Analytics, Time Series Analysis, Numerical Optimization, Hypothesis Testing, Causal Inference, Network Science, Data Science, Artificial Intelligence (AI), Numerical Methods

Tech Stack: Python, MATLAB, NumPy, Pandas, Scikit-Learn, PyTorch, BoTorch, Matplotlib, Spotfire, R,, SQL, Kubernetes, Docker, Git, CI/CD, Excel, Azure

Experience

MARCH 2022 – PRESENT

Sr Data Scientist I | Corteva Agriscience | Indianapolis, IN

- Discovered optimized recipes for testing obtained using ML techniques like adaptive experimentation, Bayesian optimization and reinforcement learning that resulted in \$10 million annual savings for the organization.
- Developed an internal production-ready ML and AI software platform in collaboration with a small team.
- Used Generative AI (GenAI), including Natural Language Processing (NLP) techniques, Large Language Models (LLM), and retrieval-augmented generation (RAG) to create a chatbot that summarizes internal R&D data for researchers.
- Employed statistical techniques such as linear mixed modeling and hypothesis testing to isolate effects of certain variables in manufacturing processes.
- Provided statistical and data science support to process optimization at multiple points in the R&D scale-up pipeline.

JANUARY 2021 – MARCH 2022

Data Scientist | Self-directed | Chicago, IL

- Applied modern data science and ML techniques to data sets with thousands of entries in health, economics, marketing, and physics to gain insights.
- Worked with simple neural networks, convolutional neural networks (CNN), recurrent neural networks (RNN), gradient boosted trees, and random forest (RF).

JANUARY 2017 – DECEMBER 2021

Graduate Student Researcher | Georgia Institute of Technology School of Physics | Atlanta, GA

- Performed statistical analysis, including hypothesis testing, on distributions of fluid flow pattern features.
- Predicted the evolution of fluid flows using convolutional neural networks and reservoir computing for up to 10 Lyapunov time units.
- Developed a technique to detect a specific sequence of flow snapshots in a turbulent flow over hundreds of time steps.
- Used persistent homology to detect topological signatures in very large (105 time steps) time-series of flow patterns consisting of 106 pixels
- Designed an experiment to detect surface waves on a centimeter-wide fluid flow using a shadowgraph imaging technique

Education

DECEMBER 2021

PhD, Physics | President's Fellow | Georgia Institute of Technology School of Physics | Atlanta, GA

Thesis title: *Investigation of spatiotemporal chaos using persistent homology*

MAY 2016

Bachelor's, Physics and Mathematics | Highest Honors | Vanderbilt University | Nashville, TN

Thesis title: *Ps2- in a magnetic field: structure and stability in the M=0 state.*

Publications and Presentations

Tregoning, B., George-Kennedy, A., Miroslav, K., Grigoriev, R., & Schatz, M. F. (2022). Using persistent homology to detect shadowing of unstable solutions (in preparation).

Tregoning, B., Mukherjee, S., Suri, B., Mischaikow, K., Paul, M. R., & Schatz, M. F. (2022). Quantifying plume statistics in spatiotemporally chaotic Rayleigh-Bénard convection using persistent homology (under review).

Tregoning, B., & Stewart, S. G. (2014). Predicting navigational error of visual binary stars. *Naval Engineering Journal*, 126.4, 169–172.

Tregoning, B., Walker, C., & Brown, D. (2025). Adaptive experimental design for fermentation process optimization, In 2025 Spring Meeting and 21st Global Congress on Process Safety. AIChE, 2025. Dallas, TX, USA. <https://aiche.confex.com/aiche/s25/meetingapp.cgi/Paper/703343>

Schatz, M., **Tregoning, B.**, Barnett, J., Yoda, M., & Grigoriev, R. (2019). Experimental Study of Roll-Hydrothermal Wave Coexistence in Convection Driven by Buoyancy and Thermocapillarity, In 72nd Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2019), Seattle, Washington, USA. <https://meetings.aps.org/Meeting/DFD19/Session/S08.3>

Tregoning, B., Mukherjee, S., Levanger, R., Cyranka, J., Mischaikow, K., Paul, M., & Schatz, M. (2019). Characterizing Spatiotemporal Dynamics in Fluid Flows using Persistent Homology, In Invited Seminar at Los Alamos National Labs, Los Alamos, New Mexico, USA.

Tregoning, B., Mukherjee, S., Levanger, R., Xu, M., Cyranka, J., Mischaikow, K., Paul, M., & Schatz, M. (2019). Using Persistent Homology to Compare Chaotic Dynamics Between Experiments on and Simulations of Rayleigh-Bénard Convection, In 72nd Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2019), Seattle, Washington, USA. <https://meetings.aps.org/Meeting/DFD19/Session/G14.4>

Tregoning, B., Levanger, R., Cyranka, J., Mukherjee, S., Paul, M., Mischaikow, K., & Schatz, M. (2018). Using topology to identify large Lyapunov vector magnitude in Rayleigh-Bénard convection, In 71st Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2018), Atlanta, Georgia, USA. <https://meetings.aps.org/Meeting/DFD18/Session/G33.5>